

Subsystem Specification

Power

Store and distribute energy safely. Give motor power only when an adult has inserted the magnetic key. Measure what's used.

Document	BOATY-SSS-PWR
Issue	Issue D (baselined, sdr-baseline-1)
Date	30 September 2026
Status	Baselined at the System Design Review (tag sdr-baseline-1); changes only by change request
Parents	SRS Issue H, ADD Issue H, ICD Issue H, FMEA Issue G, KCL Issue D
Content	10 allocated SRS requirements → 22 subsystem requirements (21 M, 1 S, 0 C)

Document control

Issue	Date	Change	By
A	28 September 2026	First issue, for review.	Claude (drafted)
B	28 September 2026	CR-03: tested salvaged 18650 cells (ADD DD-18) with acceptance tests PWR-D20 and records PWR-D21.	Claude, owner decision
C	29 September 2026	Key Component List and CR-04: the key switch is a high-side P-MOSFET (PWR-D07); supplies must tolerate pack sag near empty (PWR-D22).	Claude, owner decision
D	30 September 2026	SDR RID-12: power budget docs/budgets/power.csv now exists, generated from the ADD loads and the parameter baseline, and checked by the build. PWR-D18 corrected to 20 Wh: the salvaged-cell acceptance limit (2500 mAh, CR-03) gives 21.6 Wh; endurance 117 min against 40 min needed. SDR WP5 (RID-08): issue, parents and applicable documents taken from the baseline register; issued as the SDR baseline candidate.	Claude

Role	Name	Signature / date
Author	Claude, for Project Boaty	
Reviewer / approver	Project owner	

Review guidance: section 5 is what will be built and tested. Check that each requirement is right, sufficient and testable. Section 4 shows every parent requirement is covered (the build script checks this).

Contents

1. Scope	3
1.1 Purpose	3
1.2 Boundary	3
1.3 Breakdown	3
2. Applicable documents	3
3. Interfaces	3
4. Parent requirements	4
5. Subsystem requirements	5
5.1 Energy storage	5
5.2 Protection and switching	5
5.3 Conversion and sensing	5
5.4 Harness	6
5.5 Budgets and thermal	6
6. Design analysis: power budget	7
7. Design constraints and budgets	7
8. Verification and open items	7

1. Scope

1.1 Purpose

Store and distribute energy safely. Give motor power only when an adult has inserted the magnetic key. Measure what's used.

1.2 Boundary

Inside this subsystem	Outside (and who owns it)
• 3S Li-ion 18650 pack with BMS • Main fuse and main switch • Magnetic arming key, reed switch and MOSFET motor-rail switch • Power module (V/I sense) • 5 V buck for the mission computer • Harness and connectors	• Battery charger (owned; off-boat) • Helm battery monitoring configuration (HLM)

1.3 Breakdown

Item	Description
PWR-1 Battery pack	3 × tested salvaged 18650, 3S BMS, XT60
PWR-2 Protection & switching	20 A fuse, IP67 main switch, key switch
PWR-3 Conversion & sensing	Power module, 5 V 3 A buck, rail divider
PWR-4 Harness	Silicone wire, XT60/XT30, keyed plugs

2. Applicable documents

Document	Use
BOATY-SRS-001 Issue H	Parent requirements
BOATY-ADD-001 Issue H	Allocation, decisions, budgets, early verification items
BOATY-ICD-001 Issue H	Interface definitions (all listed in section 3 apply in full)
BOATY-FMEA-001 Issue G	Failure modes and actions carried into this issue
BOATY-KCL-001 Issue D	Component data, parameters and cost

3. Interfaces

The subsystem shall meet every interface below as defined in the ICD. 'Owner' means this subsystem maintains the definition.

Interface	Other end	Role	ICD
IF-06 Motor power	Propulsion	Owner	\$6
IF-07 Helm power & sensing	Helm	Owner	\$6
IF-08 Mission computer power	Mission computer	Owner	\$6
IF-18 Box & glands	Hull & structure	Conforms	\$8

4. Parent requirements

Primary (10): this subsystem is responsible for meeting and verifying these SRS requirements.

SRS	Pri	Requirement (abridged)	Met by
MOD-003	M	Arming shall require two deliberate adult actions: the arming switch on the boat set to ARM, and arming confirmed at...	PWR-D07, PWR-D08
PWR-001	M	Propulsion and avionics shall run from one removable 3S (nominal 11.1 V) lithium battery of 25 to 50 Wh.	PWR-D01, PWR-D20, PWR-D21
PWR-002	M	The battery shall be protected against over-discharge, over-current and short circuit (BMS or equivalent).	PWR-D02, PWR-D20
PWR-003	M	A main fuse shall sit within 50 mm of the battery positive, rated no more than twice the maximum continuous current.	PWR-D05
PWR-004	M	A waterproof main power switch shall be operable without opening the electronics box.	PWR-D06
PWR-005	M	Avionics shall have a separate regulated 5 V supply. A full reverse-to-forward thrust step shall not reset or brown...	PWR-D22, PWR-D10, PWR-D11, PWR-D12
PWR-006	M	Battery voltage and current shall be measured, logged and used for state of charge and failsafes.	PWR-D13
PWR-007	M	The battery shall be removed from the boat for charging, and charged only off-boat.	PWR-D03, PWR-D04
PWR-008	M	All power connectors shall be polarised and keyed, with no two same-type connectors doing different jobs.	PWR-D14, PWR-D15
PWR-011	S	A power budget shall be kept up to date alongside the design.	PWR-D17

Supporting (7): another subsystem owns these, but this subsystem must provide something for them. Where relevant, section 5 includes the contribution.

MOD-002, PRE-003, PRE-007, MEC-012, PWR-009, ENV-004, CHD-003

5. Subsystem requirements

Requirements are numbered PWR-Dnn. Pri, Ver and Stage follow SRS conventions (M/S/C; Test, Demonstration, Inspection, Analysis, Simulation; SIM, BENCH, POOL, LAKE). 'Trace' lists the parent SRS requirements, interfaces (IF), decisions (DD) and early verification items (V) it implements.

5.1 Energy storage

ID	Requirement	Pri	Ver	Stage	Trace
PWR-D01	The battery shall be 3S1P Li-ion 18650. Cells rated ≥ 10 A continuous, measured capacity ≥ 2500 mAh; 27-36 Wh nominal.	M	I	BENCH	PWR-001
PWR-D02	A 3S BMS shall provide over-discharge (2.5-2.8 V/cell), over-current (trip ≤ 25 A), short-circuit and balance protection.	M	T	BENCH	PWR-002
PWR-D03	The pack shall be in a rigid sleeve with XT60 and a balance lead, removable from the box by an adult in < 30 s without tools.	M	D	BENCH	PWR-007
PWR-D20	Salvaged cells shall be accepted only if each: has a known source and no dents, wrapper damage or leakage; holds ≥ 3.6 V after 7 days' rest from full (self-discharge ≤ 0.1 V); has a measured capacity ≥ 2500 mAh at 1 A; and has internal resistance ≤ 60 m Ω . The three cells shall match within 5% capacity and 10 m Ω . Anything else is rejected.	M	T	BENCH	PWR-001, PWR-002, DD-18, FM-23, A-20
PWR-D21	Each accepted cell shall be labelled with an ID, and its test results recorded in the repository. The pack's capacity shall be re-measured each season.	M	I	BENCH	PWR-001, A-20
PWR-D04	The boat shall have no charging port. Charging is by external balance charger only.	M	I	BENCH	PWR-007

5.2 Protection and switching

ID	Requirement	Pri	Ver	Stage	Trace
PWR-D05	A 20 A blade fuse shall sit ≤ 50 mm from the battery positive.	M	I	BENCH	PWR-003, IF-06
PWR-D06	The main switch shall be IP67, outside the box, and either rated ≥ 20 A or control a solid-state switch rated ≥ 30 A.	M	I	BENCH	PWR-004
PWR-D07	The motor rail shall be switched by the magnetic key per IF-06: reed switch driving a high-side P-MOSFET in the positive feed, off when the key is absent, with a soft-start gate. Low-side switching is not allowed (ESC current would return through signal grounds).	M	T	BENCH	MOD-003, IF-06
PWR-D22	Every supply on the pack (flight controller, ESCs, 5 V buck) shall work down to 7.5 V, the rail voltage at the critical battery threshold (9.6 V resting) under the helm's power limit (HLM-D43) with acceptance-limit cells (0.22 Ω pack). The Matek F405-TE (9 V minimum) failed this analysis, hence CR-04.	M	A	BENCH	PWR-005, DD-19
PWR-D08	Motor-rail voltage shall be divided to the helm's second voltage input so arming is refused below 9.0 V.	M	T	BENCH	MOD-003, PRE-007, V-13
PWR-D09	At power-up the motor rail shall be off unless the key is present. The helm is disarmed regardless (HLM-D04).	M	T	BENCH	MOD-002

5.3 Conversion and sensing

ID	Requirement	Pri	Ver	Stage	Trace
PWR-D10	The helm shall be powered from the main bus before the key switch, through its own regulator (IF-07).	M	I	BENCH	PWR-005, IF-07
PWR-D11	The mission computer shall have a dedicated 5.1 V ≥ 3 A buck (IF-08).	M	T	BENCH	PWR-005, IF-08

ID	Requirement	Pri	Ver	Stage	Trace
PWR-D12	Helm and mission-computer supplies shall stay in regulation through a full reverse-to-forward thrust step at 12.6 V and at 9.0 V.	M	T	BENCH	PWR-005
PWR-D13	Battery voltage and current shall be measured per IF-07 ($\pm 1\%$ V, $\pm 5\%$ I after calibration).	M	T	BENCH	PWR-006, PRE-003

5.4 Harness

ID	Requirement	Pri	Ver	Stage	Trace
PWR-D14	Connectors: XT60 battery, XT30 per ESC, JST-XH balance, a keyed 2-pin for the mission computer. No connector type used for two functions. All polarised.	M	I	BENCH	PWR-008
PWR-D15	Wire: 16 AWG battery to bus, 18 AWG to ESCs, silicone insulated. Joints soldered and heat-shrunk. Strain relief at every gland.	M	I	BENCH	PWR-008, MEC-012
PWR-D16	No exposed live conductor outside the box.	M	I	BENCH	CHD-003

5.5 Budgets and thermal

ID	Requirement	Pri	Ver	Stage	Trace
PWR-D17	A power budget (docs/budgets/power.csv) shall be kept current, with $\geq 30\%$ margin at the design endurance.	S	A	BENCH	PWR-011
PWR-D18	Usable energy ≥ 20 Wh (3S of cells accepted at ≥ 2500 mAh, 80% usable; was 25 Wh for 3.0 Ah cells before CR-03), supporting ≥ 40 min at the budgeted cruise load.	M	A	BENCH	PWR-009
PWR-D19	The battery shall sit ≥ 30 mm from the ESCs. Box internal temperature ≤ 50 °C after 60 min at 30 °C ambient in sun.	M	T	BENCH	PWR-010

6. Design analysis: power budget

Load	W
Flight controller + GNSS	0.8
Pi Zero 2W + camera + Wi-Fi (average)	1.8
LED beacon	0.3
ESC idle (2)	0.2
Motors at cruise (estimate)	8.0
Total at cruise	11.1
Usable energy (3S, 2.5 Ah accepted cells, 80%)	21.6 Wh
Endurance estimate	≈ 117 min (≥ 40 min required, ≥ 60 target)
Margin at 40 min	192%

Generated from docs/budgets/power.csv (PWR-D17), which the document build regenerates and checks.

Circuit: battery → fuse → main switch → power module → bus. The bus feeds the helm (own regulator), the 5 V buck (MCP) and, through the key-switched MOSFET, the ESCs. See ADD figure 6.

7. Design constraints and budgets

- ADD DD-09: 3S Li-ion 18650 + BMS
- ADD DD-10: magnetic key removes motor power independently of software
- Charging only off-boat (PWR-007)

Budget	Allocation
Mass allocation	≤ 260 g (estimate 250 g)
Cost allocation	£23 (salvaged cells + BMS £7, fuse/box share £2, buck £3, key switch £4, IP67 main switch £4, sensing/wiring £3)
Energy	≈ 32 Wh nominal, ≈ 26 Wh usable

8. Verification and open items

Each requirement is verified at its stage using the method shown. Test procedures are written per stage and reference these IDs.

Stage	Test	Demo	Inspect	Analysis	Sim	Reqs
SIM	-	-	-	-	-	0
BENCH	9	1	9	3	-	22
POOL	-	-	-	-	-	0
LAKE	-	-	-	-	-	0

Ref	Open item
TBC-06	Motor current measurement → fuse and MOSFET rating
TBC-07 / V-13	Arming gated on motor-rail voltage
PWR-OI-1	Closed: IP67 main switch added to the ADD BOM (Issue C)